MOBILE COMPUTING

ASSIGNMENT 3 – CO3

WIRELESS NETWORKS, MOBILE IP AND ROUTING PROTOCOLS

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Course: **Mobile Computing**

INTRODUCTION

Wireless networks allow devices to exchange information through radio links instead of depending entirely on physical cables. Mobile computing combines wireless connectivity with portable devices, applications and data processing so that users can continue working while moving. This assignment explains wireless network architecture, mobility management, Mobile IP, cellular and ad hoc networks, and routing protocols used in mobile environments.

The answers follow the seven questions in the supplied Assignment 3 handout. The report uses simple explanations, structured points and language-independent pseudocode. The handout requires both partners to understand every question, even when primary responsibilities are assigned separately.

OBJECTIVES

- Understand wireless network components and communication architecture.
- Explain mobility, handoff, packet loss and transmission delay.
- Describe Mobile IP registration, tunnelling and mobility challenges.
- Compare cellular and ad hoc wireless networks.
- Study Mobile IP applications and their routing requirements.
- Explain proactive DSDV and reactive AODV routing.
- Design and evaluate an emergency communication network.

Q1 – WIRELESS NETWORK ARCHITECTURE AND MOBILITY ANALYSIS

1.1 Introduction to wireless networks

A wireless network connects devices using radio signals. It supports communication between mobile users, access points, base stations and network services. Wireless networks are useful where users move frequently or where installing cables is difficult.

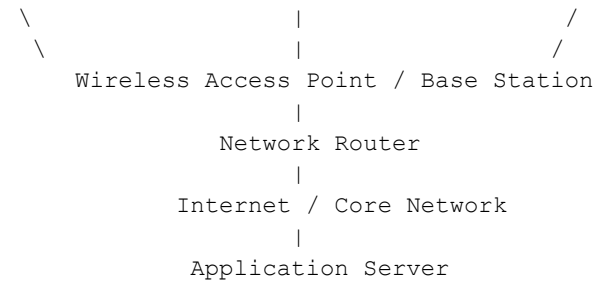
1.2 Main components

- Mobile node: A phone, tablet, laptop, ambulance terminal or other device that communicates while moving.
- Wireless access point: Connects wireless devices to a local network, commonly in a Wi-Fi environment.
- Base station: Provides radio coverage and connects mobile devices to the cellular network.
- Communication link: The wireless path that carries voice, video, messages and data packets.
- Router or gateway: Transfers packets between networks and selects the next network path.
- Network server: Provides application services, authentication, storage and other resources.

1.3 Wireless network architecture

A basic architecture can be represented as:

```
Mobile User A       Mobile User B       Mobile User C
```



Mobile nodes communicate with a nearby access point or base station. The access point forwards packets through a router to the required server or destination. In a cellular network, the operator's core network manages mobility, authentication and communication between cells.

1.4 Characteristics and limitations

- Mobility: Users can change their location while remaining connected.
- Limited bandwidth: Wireless spectrum is shared, so available capacity may be lower than a dedicated wired link.
- Variable signal strength: Distance, walls, interference and obstacles affect signal quality.
- Connectivity changes: A device may move out of coverage or switch between access points.
- Security: Wireless communication needs authentication and encryption to prevent unauthorised access.
- Power usage: Continuous radio transmission and scanning can consume battery energy.

1.5 Effect of mobility on performance

When a mobile node moves away from its current access point, signal strength may decrease. The device may need to connect to another access point or base station. This process is called handoff or handover. During handoff, packets may be delayed, lost or delivered out of order. Fast movement and overlapping coverage can increase the number of handoffs.

1.6 Wireless mobility scenario

Consider a student walking from Building A to Building B while attending an online class. Initially, the phone is connected to Access Point 1. As the student moves, the signal from Access Point 1 becomes weak. The phone detects Access Point 2, authenticates and changes its point of attachment.

```
ALGORITHM PerformWirelessHandoff(MobileNode, OldAP, NewAP)

INPUT:
    MobileNode, OldAP, NewAP, minimum signal threshold
OUTPUT:
    Handoff status

BEGIN
    Measure signal quality of OldAP and NewAP
    IF SignalQuality(NewAP) < MinimumThreshold THEN
        RETURN HANDOFF_NOT_REQUIRED
    END IF
```

```
    Authenticate MobileNode with NewAP
    IF Authentication = SUCCESS THEN
        Update point of attachment to NewAP
        Update network and routing information
        Resume packet transmission
        RETURN HANDOFF_SUCCESS
    ELSE
        Keep OldAP if its link is still available
        RETURN HANDOFF_FAILURE
    END IF
END
```

1.7 Performance measures

The scenario can be evaluated using packet delivery ratio, packet loss, average delay and number of handoffs. Packet delivery ratio is the number of packets received divided by the number of packets sent, multiplied by 100. Average delay is the total packet delivery time divided by the number of packets received.

Expected behaviour: at low mobility, the connection is more stable and packet delivery is usually higher. At high mobility, handoffs occur more frequently and the possibility of delay and packet loss increases. Actual numerical values must be collected from the simulation or network test.

1.8 Q1 conclusion

Wireless networks provide flexible communication for mobile users, but their performance depends on bandwidth, signal strength, interference and mobility. A suitable handoff procedure helps maintain communication when a user moves between coverage areas.

Q2 – MOBILE IP AND MOBILITY MANAGEMENT

2.1 Introduction to Mobile IP

Mobile IP allows a mobile node to move from one network to another while keeping its permanent home IP address. It supports continued communication even when the device changes its network attachment.

2.2 Main entities

- Mobile Node (MN): The moving device that needs continuous network connectivity.
- Home Agent (HA): A router in the home network that stores the mobile node's current location information and forwards packets toward it.
- Foreign Agent (FA): In traditional Mobile IPv4, a router in the visited network that assists the mobile node with registration and packet delivery.
- Correspondent Node (CN): The device or server communicating with the mobile node.
- Home Address: The permanent IP address assigned to the mobile node in its home network.
- Care-of Address (CoA): The temporary address representing the mobile node's current location.

2.3 Mobile IP architecture and operation

```
Correspondent Node
       |
       | Packet addressed to Home Address
       v
    Home Agent
       |
       | Encapsulated packet / tunnel
       v
 Foreign Agent or Care-of Address
       |
       v
   Mobile Node
```

When the mobile node enters a foreign network, it discovers a mobility agent and obtains a care-of address. It sends a registration request to the home agent. After successful registration, the home agent creates a binding between the home address and care-of address. Packets sent by the correspondent node reach the home agent, which tunnels them to the mobile node's current location.

2.4 Registration and location update

```
ALGORITHM RegisterMobileNode(MN, HA, CoA)

INPUT:
    Mobile Node, Home Agent, Care-of Address
OUTPUT:
    Registration result

BEGIN
    Create Registration Request
    Include MN.HomeAddress and CoA
    Authenticate the request
    SEND Registration Request TO HA
    WAIT FOR Registration Reply

    IF Reply.Status = ACCEPTED THEN
        Binding[MN.HomeAddress] ← CoA
        RETURN REGISTRATION_SUCCESS
    ELSE
        RETURN REGISTRATION_FAILURE
    END IF
END
```

2.5 Packet delivery and tunnelling

The home agent intercepts packets addressed to the mobile node's home address. It encapsulates the original packet inside another packet and sends it to the care-of address. The tunnel endpoint removes the outer header and delivers the original packet to the mobile node.

2.6 Mobile IP challenges

- Registration delay: The mobile node must discover the new network and register its care-of address.

- Tunnelling overhead: Encapsulation adds an extra IP header and consumes bandwidth.
- Triangular routing: Packets may travel from the correspondent node to the home agent and then to the mobile node, creating a longer path.
- Packet loss: Packets may be lost while the mobile node is changing networks or updating registration.
- Security: Registration messages must be authenticated to prevent false location updates.
- Scalability: A large number of mobile nodes can create additional registration and forwarding load.

2.7 Joint design: frequent network changes

Frequent movement can cause repeated registration, additional signalling, temporary packet loss and increased delay. These problems can be reduced by using fast agent discovery, efficient registration, buffering packets during handoff, reliable authentication and suitable route optimisation. The device should avoid unnecessary handoffs by checking signal quality and using a suitable threshold.

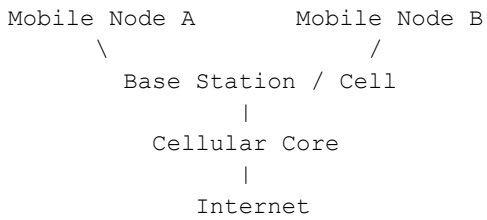
2.8 Q2 conclusion

Mobile IP supports location-independent communication by maintaining a binding between a permanent home address and a temporary care-of address. Registration and tunnelling help deliver packets to a moving device, but delay, overhead, security and routing efficiency must be considered.

Q3 – CELLULAR NETWORKS VERSUS AD HOC WIRELESS NETWORKS

3.1 Cellular network

A cellular network divides a geographical area into smaller regions called cells. Each cell is served by a base station. Mobile devices communicate through the base station, which connects them to the operator's core network. When a user moves from one cell to another, the network performs handoff.

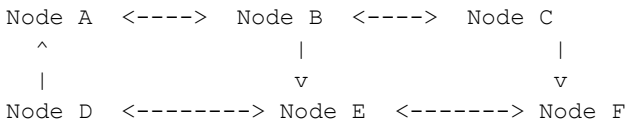


Cellular networks provide wide-area coverage, centralised management and support for a large number of users. They depend on base stations and supporting infrastructure. If the infrastructure fails, service in the affected area may be interrupted.

3.2 Ad hoc wireless network

An ad hoc network is formed directly between wireless nodes without requiring fixed access points or base stations. Each node can act as both a host and a router. If the destination is outside

direct radio range, packets are forwarded through intermediate nodes using multi-hop communication.



Ad hoc networks are useful for emergency response, disaster recovery, temporary events and field operations. Their topology changes when nodes move, so routes must be discovered and maintained dynamically.

3.3 Main differences

- Infrastructure: Cellular networks use base stations and core infrastructure; ad hoc networks can operate without fixed infrastructure.
- Communication: Cellular devices usually communicate through base stations; ad hoc nodes can communicate directly or through multiple peer nodes.
- Routing: Cellular networks use infrastructure-assisted routing; ad hoc networks use distributed routing.
- Topology: Cellular cell structure is planned; ad hoc topology changes dynamically.
- Mobility: Cellular systems manage handoff between cells; ad hoc systems update routes as links and neighbours change.
- Scalability: Cellular networks are designed for wide-area service; ad hoc scalability depends on node density, routing overhead and radio range.
- Failure handling: Cellular infrastructure failure can affect a whole area; ad hoc networks may find alternate peer routes if enough nodes remain connected.

3.4 Emergency/disaster decision

For a disaster area where fixed infrastructure is unavailable, an ad hoc network is the more suitable initial choice. Rescue teams, police vehicles and ambulances can form a temporary network and forward messages through nearby nodes. If some cellular infrastructure remains available, a hybrid design can connect the ad hoc network to the wider network through a functioning gateway.

3.5 Q3 conclusion

Cellular networks are suitable for planned, wide-area communication with available infrastructure. Ad hoc networks are suitable for rapidly deployed communication where infrastructure is damaged or unavailable. The selection depends on coverage, mobility, deployment time and infrastructure availability.

Q4 – APPLICATIONS OF MOBILE IP AND ROUTING REQUIREMENTS

4.1 Selected application: emergency vehicle communication

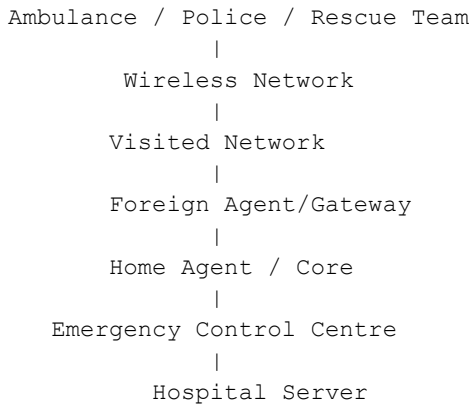
Emergency vehicle communication is a suitable Mobile IP application because ambulances, police vehicles and rescue teams move between geographical areas while exchanging important

information. The system should support location updates, continuous communication, reliable packet delivery and secure access.

4.2 Network entities

- Ambulance terminal: Sends patient status, location and emergency reports.
- Police vehicle terminal: Shares road closures, traffic information and safety alerts.
- Rescue team device: Sends field reports and requests support.
- Hospital server: Receives patient information and prepares medical support.
- Home Agent: Maintains the mobile node's home-address binding.
- Foreign Agent or visited-network gateway: Supports access in the visited network.
- Correspondent Node: The hospital or emergency control server communicating with the mobile unit.

4.3 Application architecture



4.4 Communication flow

1. The emergency vehicle connects to the available wireless network.
2. The mobile node discovers the visited network and obtains a care-of address.
3. The mobile node registers its current location with the home agent.
4. The ambulance sends patient and location information to the emergency server.
5. The home agent forwards packets through the registered tunnel when required.
6. When the vehicle moves to another network, it updates its registration and continues communication.

4.5 Routing and performance requirements

- Low delay: Emergency alerts and patient information should reach the control centre quickly.
- High packet delivery: Critical information should be delivered reliably.
- Mobility support: The connection should continue while vehicles change networks.
- Security: Authentication and encryption should protect patient and operational information.
- Reliability: The system should detect link failure and use an alternate connection when available.
- Scalability: The network should support many emergency vehicles and field devices.

4.6 Suitability of Mobile IP

Mobile IP is useful when a mobile node must retain its home address while changing networks. It can support continuous access to emergency services, but registration delay and tunnelling overhead must be managed. In a disaster where all fixed infrastructure is unavailable, Mobile IP alone is not enough; an ad hoc routing network is also needed for local communication.

4.7 Q4 conclusion

Mobile IP can support emergency communication by maintaining a mobile node's reachability across networks. The application should combine mobility management with reliable routing, security and failure handling.

Q5 – PROACTIVE ROUTING PROTOCOLS IN MOBILE NETWORKS

5.1 Proactive routing

A proactive routing protocol maintains routes to destinations before data transmission is requested. Each node periodically exchanges routing information and updates its routing table. This reduces route discovery delay when an application needs to send data, but periodic updates create control overhead.

5.2 DSDV protocol

Destination-Sequenced Distance Vector (DSDV) is a proactive routing protocol based on distance-vector routing. It uses destination sequence numbers to identify fresh route information and reduce routing loops. Each routing entry stores the destination, next hop, hop count and sequence number.

5.3 DSDV operation

7. Each node maintains a routing table containing known destinations.
8. Nodes periodically broadcast routing updates to neighbours.
9. A receiving node checks the sequence number and route metric.
10. A newer sequence number is preferred because it represents fresher information.
11. If sequence numbers are equal, the route with the smaller hop count can be selected.
12. Expired or invalid routes are removed or marked unavailable.

5.4 DSDV route-update pseudocode

```
ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)

INPUT:
    Destination, NextHop, HopCount, SequenceNumber
OUTPUT:
    Updated routing table

BEGIN
    NewHopCount ← ReceivedRoute.HopCount + 1
    NewSequence ← ReceivedRoute.SequenceNumber
```

```
    IF Destination is not in RouteTable THEN
        ADD ReceivedRoute to RouteTable
    ELSE IF NewSequence > RouteTable[Destination].SequenceNumber THEN
        REPLACE RouteTable[Destination]
    ELSE IF NewSequence = RouteTable[Destination].SequenceNumber AND
            NewHopCount < RouteTable[Destination].HopCount THEN
        REPLACE RouteTable[Destination]
    END IF

    RETURN RouteTable
END
```

5.5 Example topology and route change

Consider nodes A, B, C and D. Initially, A reaches D through B and C. If the link between B and C fails, routing updates inform A that the previous route is no longer valid. A new route may be selected if another path exists. Sequence numbers help nodes distinguish fresh route information from old information.

```
Before failure:
A ---- B ---- C ---- D


After B-C link failure:
A ---- B      C ---- D
 \_____________________/
       Alternate path
```

5.6 Performance analysis

The main advantage of DSDV is that routes are already available, so initial route lookup delay is low. Its disadvantage is periodic routing-control traffic, even when there is little user data. In larger networks, more route information must be maintained and exchanged. Frequent mobility can cause repeated updates and convergence delay.

5.7 Q5 conclusion

DSDV is useful when routes should be available before data transmission and the network topology does not change too rapidly. Its sequence numbers improve route freshness, but periodic updates consume bandwidth and energy.

Q6 – REACTIVE ROUTING AND PROTOCOL COMPARISON

6.1 Reactive routing

A reactive routing protocol discovers a route only when a source node needs to send data and no valid route is already available. It reduces periodic control traffic, but the first packet may experience route discovery delay.

6.2 AODV protocol

Ad hoc On-Demand Distance Vector (AODV) is a reactive routing protocol. It uses Route Request (RREQ), Route Reply (RREP) and Route Error (RERR) messages to discover and maintain routes.

- RREQ: Broadcast by a source node to discover a route to a destination.
- RREP: Sent back when the destination or a node with a valid route is found.
- RERR: Notifies affected nodes when a route or link becomes unavailable.

6.3 AODV route discovery

```
ALGORITHM ReactiveRouteDiscovery(Source, Destination)

INPUT:
    Source and Destination
OUTPUT:
    Valid route or ROUTE_NOT_FOUND

BEGIN
    IF RouteTable[Destination].State = VALID THEN
        RETURN RouteTable[Destination]
    END IF

    Create Route Request with a new Request ID
    BROADCAST RREQ to neighbours
    WAIT for RREP or route failure

    IF RREP is received THEN
        Establish reverse and forward paths
        Mark destination route as VALID
        RETURN RouteTable[Destination]
    ELSE
        RETURN ROUTE_NOT_FOUND
    END IF
END
```

6.4 AODV route maintenance

AODV monitors active routes. If a link breaks, the affected route is marked invalid. If an alternate valid route exists, it can be used. Otherwise, a Route Error message is sent to the source. If communication is still required, a new route discovery process begins.

6.5 DSDV versus AODV

DSDV maintains routes continuously, while AODV discovers routes on demand. DSDV usually provides lower initial route setup delay because routes are already stored. AODV can reduce control overhead when traffic is infrequent because it does not continuously advertise all routes. However, AODV introduces route discovery delay when a route is not available.

For frequent data traffic and a relatively stable topology, DSDV can be suitable. For intermittent communication and limited bandwidth, AODV can be more efficient. Under frequent topology

changes, AODV may avoid some periodic update overhead, but repeated route breaks can trigger repeated discovery and delay.

6.6 Joint decision for a highly dynamic network

For a highly dynamic mobile network with frequent topology changes and intermittent traffic, AODV is a reasonable selection because it discovers routes when needed and avoids continuous full-table updates. If the application requires very low first-packet delay and sends data continuously, DSDV may be preferable. The final choice should be supported by measured delay, packet delivery and routing overhead.

6.7 Q6 conclusion

AODV is demand-driven and uses RREQ, RREP and RERR messages. DSDV is proactive and maintains routes using sequence numbers. The better protocol depends on traffic frequency, mobility, bandwidth and delay requirements.

Q7 – DESIGN A MOBILE EMERGENCY COMMUNICATION NETWORK

7.1 Problem description

During a natural disaster, ambulances, police vehicles, rescue teams and medical staff move continuously. Fixed communication infrastructure may be damaged or unavailable. The network must support communication with minimum delay, reliable packet delivery and the ability to recover from link or node failures.

7.2 Student A – independent design proposal

Student A proposes a wireless architecture that uses mobile nodes and available cellular or Wi-Fi access where possible. When a fixed network is available, Mobile IP maintains reachability as vehicles move between networks. When infrastructure is unavailable, nearby mobile nodes form an ad hoc network. The design includes handoff management, location updates and packet buffering during short disconnections.

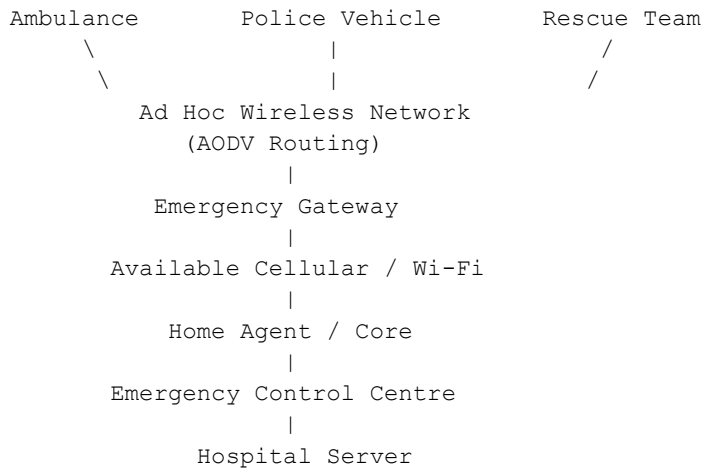
7.3 Student B – independent design proposal

Student B proposes a reactive routing protocol such as AODV for the ad hoc portion of the network. AODV discovers routes when communication is required and repairs routes after link failure. The design monitors link status, sends route error messages and initiates new route discovery when an alternate path is unavailable. Performance is evaluated using delay, packet delivery ratio and routing overhead.

7.4 Comparison and joint final design

Both proposals support mobile communication and failure handling. Student A focuses on network architecture and mobility management, while Student B focuses on routing and route maintenance. The joint design combines both approaches: Mobile IP supports communication across available IP networks, and AODV supports multi-hop communication among mobile nodes when fixed infrastructure is unavailable.

7.5 Proposed architecture



The emergency gateway connects the local ad hoc network to any available external network. If the external network fails, local communication can continue among reachable ad hoc nodes. The gateway can store messages and forward them when connectivity returns.

7.6 Mobile IP registration and packet delivery

13. A mobile node detects the available network and obtains a care-of address when it enters a visited network.
14. The mobile node sends a registration request to its home agent.
15. The home agent authenticates the request and updates the home-address binding.
16. Packets addressed to the mobile node are tunnelled to its care-of address.
17. If the mobile node moves again, it updates its registration with the new location.

7.7 Selected routing protocol and justification

AODV is selected for the ad hoc part because the network topology changes as vehicles and rescue teams move. It discovers routes when data must be sent and can repair a route after a link break. It avoids continuous routing-table advertisements when traffic is low. However, route discovery may add delay, so urgent traffic should be prioritised and critical routes should be monitored.

7.8 Failure handling

- Link failure: Mark the route invalid, search for an alternate route and send a route error if necessary.
- Node failure: Neighbouring nodes detect missing responses and remove stale route information.
- Network disconnection: Store critical messages locally and forward them when a connection returns.
- Gateway failure: Select another available gateway if one exists.
- Packet loss: Use acknowledgements or retransmission for critical messages.
- Security failure: Reject unauthenticated devices and protect emergency information using encryption.

7.9 Performance evaluation plan

The network should be tested with different numbers of mobile nodes and different mobility conditions. Compare AODV with DSDV using the same topology, traffic pattern and simulation duration. Record average end-to-end delay, packet delivery ratio, packet loss, routing-control overhead and route recovery time.

- Test 1: Small network with low mobility.
- Test 2: Larger network with low mobility.
- Test 3: Small network with high mobility.
- Test 4: Larger network with high mobility.
- Test 5: Temporary network disconnection followed by reconnection.

No numerical experimental results are supplied in the assignment handout. The values must be obtained by running the selected simulator or test environment. Results should not be invented.

7.10 Joint evaluation

The proposed hybrid network is suitable because it can use fixed infrastructure when available and continue local communication through ad hoc routing when infrastructure is unavailable. Mobile IP helps maintain reachability across IP networks, while AODV discovers multi-hop routes within the temporary network. The design introduces registration and routing overhead, and performance may decrease when node density is low, mobility is extremely high or the network becomes partitioned.

7.11 Final conclusion

Before selecting Mobile IP and a routing protocol, a network designer should analyse infrastructure availability, number of mobile nodes, mobility speed, traffic volume, bandwidth, delay requirements, packet delivery, security, power consumption and failure conditions. The protocol should be selected based on the actual network requirements and validated through experiments.

OVERALL CONCLUSION

This assignment examined wireless network architecture, mobility management, Mobile IP, cellular and ad hoc networks, and proactive and reactive routing. Mobile IP maintains communication across network changes using registration and tunnelling. Cellular networks provide wide-area infrastructure-based communication, while ad hoc networks support temporary communication without fixed infrastructure. DSDV maintains routes proactively, whereas AODV discovers routes on demand.

For disaster environments, a hybrid design using Mobile IP when infrastructure is available and AODV-based ad hoc routing for local communication can provide flexibility. The final design must be tested using delay, packet delivery, packet loss and routing overhead.

APPENDIX A – IMPORTANT PERFORMANCE FORMULAS

- Packet Delivery Ratio (PDR) = (Packets Received / Packets Sent) × 100.

- Packet Loss = Packets Sent − Packets Received.
- Packet Loss Percentage = (Packets Lost / Packets Sent) × 100.
- Average End-to-End Delay = Total delay of received packets / Number of received packets.
- Routing Overhead = Number of routing-control packets transmitted during the experiment.
- Throughput = Total successfully received data bits / Total simulation time.

APPENDIX B – PSEUDOCODE WRITING GUIDELINES

The supplied handout specifies that Mobile Computing pseudocode must be language-independent and must describe network decisions clearly. It should not be ordinary prose or programming code.

- Use ALGORITHM, INPUT, OUTPUT, BEGIN and END.
- Use ← for assignment and IF, ELSE, FOR and WHILE for decisions and loops.
- Use network operations such as SEND, RECEIVE, FORWARD, DELIVER, DROP and UPDATE ROUTE.
- Define important state such as HomeAddress, CareOfAddress, NextHop, SequenceNumber, RouteState and LinkState.
- Include failure cases, route discovery, registration, handoff and termination conditions.
- Make the algorithm manually traceable using a small network example.

APPENDIX C – PAIR-WORK CONTRIBUTION RECORD

The assignment handout expects both students to participate in every question. The primary responsibilities can be recorded as follows:

- Q1: Student A – network architecture; Student B – mobility analysis; joint work – wireless network analysis.
- Q2: Student A – Mobile IP operation; Student B – Mobile IP challenges; joint work – Mobile IP architecture.
- Q3: Student A – cellular network; Student B – ad hoc network; joint work – comparative analysis.
- Q4: Student A – performance analysis; Student B – application architecture; joint work – application evaluation.
- Q5: Student A – protocol design; Student B – performance investigation; joint work – proactive routing.
- Q6: Student A – reactive routing; Student B – DSDV/AODV comparison; joint work – routing evaluation.
- Q7: Student A – analytical validation; Student B – experimental validation; joint work – final network design.

Both students should be prepared to explain the complete report during the faculty verification or viva. A student should understand the partner's section and be able to explain how a change in mobility, traffic or infrastructure affects the network.

REFERENCES

18. ITA03-MP, Mobile Computing, Assignment 3 – CO3, supplied course handout.
19. ITA03-MP, Standard for Algorithmic Pseudocode, supplied with the assignment handout.
20. Perkins, C. E., Mobile IP: Design Principles and Practice, Addison-Wesley.
21. Perkins, C. E., Belding-Royer, E. and Das, S., Ad hoc On-Demand Distance Vector (AODV) Routing, RFC 3561.
22. Perkins, C. E. and Bhagwat, P., Highly Dynamic Destination-Sequenced Distance-Vector Routing (DSDV) for Mobile Computers.